Question ID: 5ab9903f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

The cost to rent a bus from Company X is $\$950$ for the first 3 hours and an additional $\$50$ per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is $\$1{,}150$, which equation represents this situation?

Answer

A. $950(t-3)+50t=1{,}150$

B. $950(3t)+50t=1{,}150$

C. $950+50(t-3)=1{,}150$

D. $950+50(3t)=1{,}150$

Correct Answer: C

Rationale

Choice C is correct. It's given that the cost to rent a bus is $\$950$ for the first 3 hours and an additional $\$50$ per hour for each hour after the first 3 hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first 3 hours can be represented by the expression $t-3$. The cost for these additional hours is $\$50$ per hour, so the cost for the additional hours can be represented by the expression $50(t-3)$. The total cost can be calculated by adding the cost for the first 3 hours to the cost for the additional hours and can be represented by the expression $950+50(t-3)$. It's also given that the total cost to rent the bus from the company for t hours is $\$1{,}150$. Thus, the equation that represents this situation is $950+50(t-3)=1{,}150$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.